



A roadmap of steganography tools: conventional to modern

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Abstract Steganography emerged as an effective technology for securing the data over the network. Its specificity of concealing the existence of the secret data supports its application in securing the information in the modern era. The desire of industry and support from various governments motivated the researchers to develop steganography tools. These spatial and transform domain tools implement different steganography techniques either solo or as a hybrid using a wide range of media as a cover file for hiding various types of data. In this paper, a systematic study of the steganography tools developed in the last three decades has been done. The comparative analysis of these tools based on specified parameters represents their strengths, limitations, applicability, and scope for future work as well. OpenPuff steganography tool spawns a huge acceptance by academics and professionals. This paper also analyze the performance of the OpenPuff tool on some unexplored parameters to validate and justify its performance.

Keywords Watermarking Tools · Spatial Domain · Transform Domain · Carrier files · Performance Analysis

1 Introduction

In today's digital world steganography has made an environment of corporative observance that has produced several valuable applications. Steganography is actively used in the deployment of a secure approach for communication in the majority of popular arenas that includes copyright production of resources, improving the robustness of image search engines, Smart IDs (where persons' information is concealed in their photographs), Audio–video management, safe exchange of secret information for firms', TV broadcasting, TCP/IP (Transmission Control Protocol/Internet Protocol) messages along with checksum concealing, safe military information transmission, and Fingerprinting applications [1].

Steganography preserves information integrity i.e. there will be no alteration in data during transmission. It is also useful for information validation which means to validate if the information is coming from the legal person and will be communicated to an intentional receiver. Along with this, it retains the privacy and confidentiality of the information [2]. Modern applications of steganography include Health Imaging Structures where a separation is essential for privacy among patients' image information or DNA (Deoxyribonucleic Acid) structures and their descriptions, e.g., physician of patient, name of the patient, address, and other details of the patient. A connection is essential to be maintained among image information and personal information of the patient. So, concealing the patient's information in an image can be a beneficial protection measure and supports in resolving such difficulties [3].

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The location of secret information in the carrier file may vary. The secret information can be spread uniformly over the entire carrier file which is known as spatial domain steganography or can be spread into a complex region where it is very challenging to identify small alterations that are known as transform-domain steganography. A complex region is also known as a region of high frequency in which there are small variations in color intensity [4]. One of the widely used spatial domain steganography is LSB (Least Significant Bit), it works by substituting the least significant bits of a carrier pixel. These alterations might be inferred as random noise, which must not have any noticeable effect on the carrier file. In the case of a 24-bit bitmap, LSB substitution doesn't cause major quality degradation. Some algorithms substitute LSB pixels of the carrier arbitrarily others change pixels in specific regions of the carrier, or some just increment or decrement pixel value of the carrier [5]. Some of the characteristic tools of LSB steganography are: S-Tools, StegCure, StegoDos, White Noise Storm, Hide & Seek, OpenStego, Hide4PGP, Silenteeye, Hide'N'Send, and Secure Engine Professional. Mentioned tools can take many file formats including BMP (Bitmap Image), GIF (Graphics Interchange Format), PNG (Portable Network Graphics), TIFF (Tagged Image File Format), JPG (Joint Photograph Group), and WAV (Waveform Audio), etc. as carrier for concealing of secret information [6].

The DCT (Discrete Cosine Transform) technique is frequently used to transform domain steganography for communicating a waveform as a weighted sum of cosines. Secret information is concealed in the cover image by changing the DCT coefficient of the cover image. DCT coefficients having a low threshold are replaced by bits of secret information. The inverse transform would generate a stego image and extract the DCT coefficient with a low threshold. The transform domain steganography contains: JSteg, OpenPuff, F3, F4 and F5, JSteg shell 2.0, JPHS, F5, outguess rebirth, Steganos Security Suite 6.0 algorithms tools, etc. [7].

Many steganography techniques and tools have been developed for concealing secret information in various digital carrier media. In this paper, analysis of various features for these tools e.g. type of carrier files, type of concealed information, the technique used, encryption, compression, password protection, etc. are presented. Along with this, to show the efficiency and reliability of the OpenPuff tool, worked on various features like concealing time, retrieval time, the security of concealed information, concealing capacity, histogram analysis, visual inspection analysis, etc.

2 Steganography tools

Steganography means concealing secret information in a digital carrier such that a hacker cannot detect whether something is concealed or not since there are negligible alterations in carrier digital media. Some steganography software tools are available to ensure that information remains a secret even in case of an attack by an unauthorized person. Significant work has already been done by researchers related to steganography tools using different digital carriers, domain, GUI (Graphical User Interface) and CMD. The categorization of these tools is given in Fig. 1.

2.1 Categorization of steganography tools based on carrier file

The steganography tools have been categorized based on the type of digital carrier media used for concealing secret information as explained:

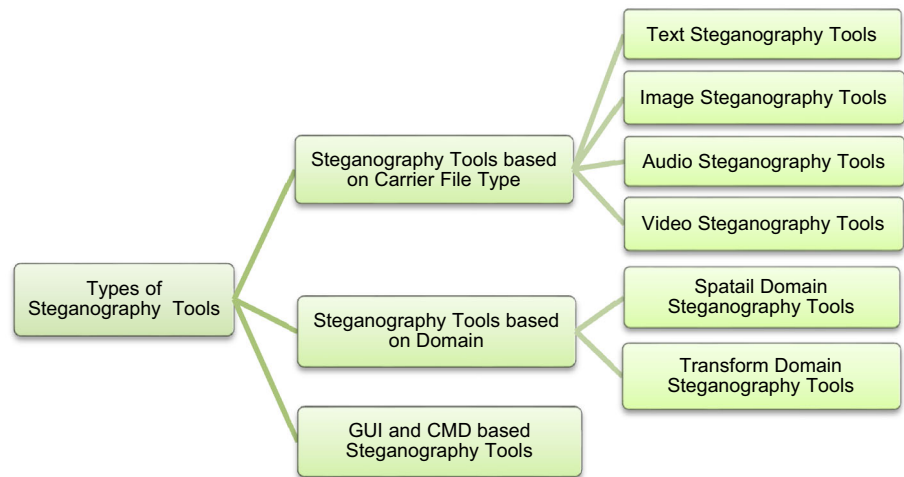
2.1.1 Text steganography tools

Text steganography tools are simplest with less information concealing capacity and low imperceptibility [8]. These tools use text as a carrier file and conceal text information. These tools provide a double level of security when a secret key is used for concealing secret information. These tools cause problems in rewriting, modifying line-height, addition, or removal of words.

Banerjee, Indradip, Souvik Bhattacharyya, and Gautam Sanyal in paper [8], has given a comprehensive review of text steganography tools. The author also reviewed various text steganalysis tools to detect the concealed secret information in the form of text. Authors in [9], presented a steganalysis attack on PDF (Portable Document Format) document against the OpenPuff steganography tool. OpenPuff steganography tool is open source software with a large user base for concealing multi-format. They detected the presence of a concealed secret PDF format with accuracy.

2.1.2 Image steganography tools

Image steganography tools are much more powerful than text steganography tools in terms of security and information concealing capacity. These tools use an image as a carrier file and can conceal text or image both. In image steganography concealed information does not stand out by making a slight change to the color values of the image. For better security of concealed information usually, more padding is done around secret information.

Fig. 1 Categorization of steganography tools

Michaud [10] has presented a review and analyzed some freeware image steganography tools. They also provided information about various digital multimedia files along with the analysis of Snow and Steghide tools in terms of their applications in various fields. Hera and Hassan Hajj-diab [11], compared various image steganography tools based on different techniques used to conceal some secret information. They calculated the PSNR (Peak Signal to noise ratio) value of stego images generated by these tools.

In paper [12], Das and Resul have presented an image steganography approach and tools which are used to conceal secret information in different carrier images. The authors have examined techniques used by a particular tool to conceal secret information along with types of concealed information and carrier files. Lakdawala, Jensi, Jiya Rankawat [13], classified image steganography and compared many image steganography tools by taking different image formats. The objective of this study was to find the most reliable and best tool available in the market for image steganography. Ammar Almomani, and Putra Sumari [14], provided the difference between cryptography, watermarking, and steganography. Image steganography for concealing secret information was also discussed. Recent steganography tools were analyzed and demonstrated for concealing secret information into images. Comparisons among many image steganography techniques used by these tools were provided. In paper [7] authors have concealed secret information into different file formats to check the robustness of image steganography tools. In the paper, they also presented experimental outcomes through various digital media files followed by the final decision whether the experimental result could reject the null hypothesis or not.

WK Lai and Z Alireza [15], explained the enhanced LSB insertion system from the simplest and straight 1-LSB to the color cycle technique. Several existing

steganography techniques were discussed and some essential complications were highlighted. In comparison to the other information concealing tools, StegCure is user-friendly with interactive GUI and cohesive navigation competencies. StegCure also uses the PKI (Public Key Infrastructure) mechanism to provide a double level of security at source and receiver sites. It conceals secret information in images using the LSB technique. I Broda and Martin [16], proposed steganalysis techniques for the detection of secret information that was grounded on the extraction of statistical properties from carrier and stego images in JPEG image format together with standardization technique. They used Support Vector Machines (SVM) classification or Bayes classifier for the preparation of the model that was later used by the same steganalysis to recognize among carrier and stego images. They compared the detection accuracy of models for these two classifiers: Support Vector Machines and Bayes classifier. They used steganography tools (nsF5, Model-Based 1, Model-Based 2, Modulo Histogram Fitting with Dead Zone, and Perturbed Quantization Steganographic) to conceal secret information.

2.1.3 Audio steganography tools

Audio steganography tools provide more security and a large amount of information concealed. These tools use inaudible frequencies of the audio file as carriers and can conceal text/image/audio information.

Yalinne, and Ben Khodja [17], developed an audio steganography tool that implements a procedure for concealing encrypted secret information in an audio file. Encryption of secret information before concealing provides a double level of security. Andrew H. Sung, and Qingzhong Liu [18], worked on some MP3 audio steganalysis technique to detect the concealed secret

information. They proposed a comprehensive method to MP3 steganalysis through developing a group of features from quantized MDCT (Multidetector DCT) coefficients. They designed frequency-based sub-band moment statistical features, accumulative Markov transition features, and accumulative neighboring joint density features on second-order derivatives. They also modeled alteration by removing shape parameters of generalized Gaussian density from different frames. Diverse feature collection algorithms are applied to increase detection precision. Signal complication and alteration density were presented to deliver a comprehensive assessment. Experimental results showed that MP3Stego, particularly for audio streams with high signal complexities were usually much more challenging to steganalysis.

2.1.4 Video steganography tools

Video steganography tools have the option of concealing secret information inside inaudible frequencies and video frames as well. These tools use multiple random video frames as a carrier file and can conceal any digital media file inside. Randomly concealed secret information increase security as well as the capacity of these tools.

Thomas and Julio Hernandez-Castro in paper [9], examined concealing procedures and other properties of some video steganography tools. Cover files play a major role in the detection of secret information. They concluded by highlighting the poor state of current video steganography tools, that no reliable and publicly available video steganography tool can provide good security. Therefore, they inspired the steganography researchers for developing secure and reachable video steganography tools.

Andreas Westfeld [19], analyzed various transform domain steganography tools to conceal secret information in transform coefficients, which provide both imperceptibility and robustness. Transform domain steganography mostly include JSteg, F3, F4, and F5 algorithms and OpenPuff tools. Gupta and Lalit Kumar [20], examined the performance of the OpenPuff tool in relevance to many image formats like JPEG, BMP, and PNG. Stego images created through the OpenPuff tool were compared on several features of an image. Pixels and bit value alterations were analyzed using the security steganalysis tool. PSNR was also calculated for the stego image and then matched based on robustness, visual distortion, etc. aimed at protection from image processing attacks. They also calculated the size of input and output file, concealing and extraction time, etc. Kundu, Dushyant, and Amit Upreti [21] provided improved security to secret information over the web, by using some existing steganography methods and tools. A comparative study of many open-source steganography tools and their methods was presented.

Techniques have been discussed based on parameters such as their strengths, weaknesses, concealing method, time to conceal secret information, etc. The benefit of one method over others depends upon the type of application and its need.

Bansal, Deepika, and Rita Chhikara [22] designed an algorithm for nsF5, PQ, and Outguess steganography tools to provide better robustness against image processing attacks. SVM and NNP (Neural Network Pattern) recognition tools were used for the extraction of sensitive features through the DCT domain. SVM and NPR (Network Pattern Recognition) tool was also compared for their accuracy. Based on the experimental results it was found that the Outguess steganography tool is the best tool among nsF5 and PQ (Post-Quantum) as it withstood the image processing attacks. Xianfeng Zhao and Yun Cao [23], proposed an effective fingerprint recognition system against the OpenPuff video steganography tool. They aimed at attacking the videos in the formats SWF, MP4, and 3GP. For attacking purposes, they elaborated on the working mechanism of OpenPuff and analyzed each possible and doubtful field for different file formats. They compared binary streams among source and stego video, the doubtful areas were marked and used for screening video. For doubtful video, stuffing bytes in those particular fields were removed as a signature which decides whether the carrier file was innocent or not. The experiment was performed on hundreds of videos to determine the efficiency of their suggested scheme. Mishra and Aayush [24], proposed a video steganography tool named VStegNET that extracts Spatio-temporal features using 3D-CNN (Convolutional Neural Network) and micro bottleneck. The proposed tool conceals $M \times N$ (RGB) secret MP4 frames into same sized carrier MP4 frames. They had trained their tool on UCF 101 action recognition MP4 dataset and assessed its performance using quantitative metrics like PSNR, SSIM (Structure Similarity Index Ratio), etc. They have also presented the advantages of their VStegNET tool over image steganography tools. To justify the robustness of their tool they used some steganalysis tools like StegExpose, SRNET.

From the literature survey, it has been found that image steganography is the focus for many researchers. Video steganography is the second most popular tool in recent years. Video steganography is more secure, robust, and have a large concealing capacity as compared to the other steganography tools. A video has a large number of frames so secret information can be concealed into multiple frames randomly at a particular time, it provides more security and concealing capacity. Concealing information into multiple frames also improves robustness against different attacks.

2.1.5 Categorization of steganography tools based on domain

Many open-source, freeware and commercial steganography tools exist on the internet today. Some of these steganography tools offer only information concealing but a few also offer encryption of secret information before concealing it. Few of these tools can only conceal secret information behind the image, but some tools can conceal behind any digital media. The basic idea behind all these tools is almost the same: to create a new steganography tool what we need is how to conceal secret information. For the steganography process to take place both the sender and receiver are required to connect to the same software. Some of these tools work in the spatial domain and some others in the transform domain. These steganography tools are available in two categories as shown:

2.1.6 Spatial domain-based steganography tools

In the spatial domain, some of the bits in carrier files are substituted by the secret information. Some commonly used spatial domain techniques are PVD (Pixel Value differencing), RPE (Random Pixel Embedding), EBE (Edges based Information Embedding), MPD (Multi-Pixel Differencing), LSB, etc. Some of the Spatial domain steganography tools are as follow:

Xiao Steganography It is open-source software having 2.14 MB size that could be used to conceal secret information in BMP images or audio files. It is a window based and user-friendly tool. First, load a BMP image or audio to its interface and then conceal secret information in the BMP image or Wav audio. It also provides encryption and hashing. Select any one from encryption algorithms like RC4 (Rivest Cipher 4), Triple DES (Data Encryption Standard), DES, Triple DES 112, RC2, and hashing SHA (Secure Hash Algorithm), MD4 (Message Digest 2), MD2, MD5. After selecting the encryption and hashing algorithm save the target file. To decode concealed secret information from this file, use this tool again. This software will read the file and will decode the hidden file from it [25].

S-Tool It is a program by Andy Brown which can conceal secret information into GIF, BMP, and audio files. S-Tools use the LSB substitution method in the spatial domain for concealing secret information. It provides lossless compression in 24-bit color and pulse code modulation. This tool also provides encryption to secret information before concealing by using DES, IDEA (International Data Encryption Algorithm), MDC (Modification Detection Code), and Triple-DES. Audio carrier files are well suitable for information concealing because they are generally large as compared to image and text files, making it challenging to find small concealed

information. This tool uses 24-bit color images and these are not usually used on the web because they tended to stand out. The tool finds the intensity of neighbor color by using the formula of distance calculation. After the calculation of the intensity of the neighbor color, every color of the dithered image should be related to two palette records one of which would transfer the secret information. S-Tool software could moderate the number of colors in the image up to 256 and that would not visually interrupt image quality. The interface of this tool is simple to use and it provides a drag and drop system to load images. When a carrier image is dragged in; the system will recommend a programmer how much information in bytes carrier image can conceal [26].

Hide'N'Send It is window-based image steganography tools having a small size of 536.04 KB. It requires .net framework 2.0 for its working environment. It first encrypts secret information using AES, RC2, RC4, F5 algorithm, and then conceals secret information using LSB. The encryption key is developed from the user password with help of hash functions RIPEMD (RIPE Message Digest), SHA512, and MD5 [27].

OpenStego It is a small size window-based tool that provides information concealing using random LSB as well as watermarking in images. It is an open-source image steganography tool developed using the Java platform. Using this tool, steganography can be performed effectively on image files having type JPEG, JPG, BMP, GIF, and PNG, etc. Stego output is stored in the PNG file format. Watermarking delivers an ability to imprint invisible watermark on image files. It also provides encryption using AES (Advanced Encryption Standard), DES, and hashing for the development of user password key [28].

StegoShare This tool allows the concealing of large secret files inside multiple carrier images. It can conceal up to 2 GB of secret information and several carrier images up to a set of 65,536. This tool uses 128-bit encryption for a double level of security. It has an average capacity of 40% which means, 10 KB information can be concealed into a 25 KB carrier image. It works on fixed location LSB substitution that makes its stego image noticeable to most steganalysis tools [29].

Hide and Seek It practices a general methodology and is quite simple to put in images and audio files. It uses the LSB bit substitution method in the spatial domain to conceal secret information inside the image. It may produce little noise in GIF color images except a greyscale image is used. Though it has been stated that greyscale GIFs don't show any bad effects like noise related to 8-bit color images which have been used for Steganography. If carrier image size increases because of the concealing of secret information then the stego-image is resized or cut to fit the required size and it may result in loss of secret information.

These features of this tool may lead hackers to the existence of concealed secret information. Hide and Seek 1.0 for Windows have no size limitations and practices an enhanced method for concealing secret information. It uses a hash function to generate a random number for password protection.

2.1.7 Transform domain-based steganography tools

In transform domain carrier digital media is transformed from one domain to another domain for concealing secret information. Some of these techniques are DCT, DWT (Discrete Wavelet Transform), IWT (Integer Wavelet Transform), and DFT (Discrete Frequency Transform). Some of the Transform domain steganography tools are shown as follow:

OpenPuff It is an open-source transform domain-based software tool and suitable for exchanging information online. This tool support information in audio (AIFF, MP3, WAV, SUN), video (3GP, MP4, MPG, VOB), image (BMP, JPG, PCX, PNG, TGA), flash-adobe (PDF, SWF, FLV) format. It provides many layers of security in the forms of 256-bit symmetric key encryption, 256-bit symmetric key scrambled, 256-bit symmetric key whitened and adaptive nonlinear carrier bit encoded. This tool also supports hashing for generating passwords. It does not require installation and hence portable [30].

VSL 1.1 VSL is open source software written in java language. This tool conceals secret information in images using LSB and DCT technique, detect its existence, and investigate its robustness using many dissimilar adaptable methods. This is a GUI based tool, provides difficult processing which could be accomplished in batch and parallel form. Secret information can be concealed with basic LSB technique, with the Karhunen–Loeve Transform or F5 algorithm, which practices DCT transformation in JPEG images. For steganalysis, RS-Analysis and Binary Similarity Measures techniques are used with the Support Vector Machines classifier technique which notices steganography in any type of carrier file [31].

Outguess-Rebirth It is a 2.72 MB portable tool for Windows, allows concealing secret information inside the JPEG image using DCT. It is open-source software and appropriate for concealed secret information transmission. Before concealing secret information, it is encrypted with the AES algorithm, scrambled, whitened, and encoded. It uses the Outguess steganography machine, which decreases the chances of detecting concealed information. Its interface is very simple and user friendly [32].

JSteg It is a small size tool of 1 KB. This tool conceals secret information inside JPEG images in LSB using the DCT technique. This tool doesn't provide the option of

encryption and compression. Information concealed can be easily be detected by the detectors [33].

F5 Steganography It is a simple steganography tool for concealing secret information in JPEG images using the DCT (F5) algorithm. It is a Java-based free tool. This tool doesn't use metadata or comment fields of image. This tool is also able to provide password protection and quality adjustment of JPEG images. It is robust against image processing tools [34].

From literature survey it is observed that more work has been done on spatial domain techniques as compared to transform domain techniques. Spatial techniques are mainly suitable for directly substituting the gray level values of pixels by secret information resulting in contrast change of the entire carrier file. Transform domain techniques determine the magnitude and phase of the carrier file. The magnitude contains the frequency component and phase can restore the carrier file to the spatial domain. This technique allows operation on the frequency component of the carrier. High-frequency components like edges and other delicate information can be simply improved.

2.2 GUI and CMD based steganography tools

This category includes tools using random parts of the carrier file to conceal the secret information. These tools are user-friendly and some of the tools even do not require installation. These tools are less secure as compare to spatial and transform domain tools. Tools in this category are available to conceal any type of secret information in any carrier file. Some of these tools are shown as follow:

Steghide It is free software that allows concealing secret information in an image or audio file by the command-line interface. To use this tool what we require is to learn the commands for concealing information in images and audio files. After concealing secret information no change occurs in the quality of the image or audio file. For the extraction of concealed information, we need to learn another set of commands. It also provides features such as encryption of secret information before concealing using blowfish algorithm, MD5 hashing for key formation from a passphrase, pseudo-random distribution of concealed bits, and capability to conceal a CRC32 (Cyclic Redundancy Checksum 32) checksum of plain information to authenticate integrity during extraction [35].

SteganPEG It is a 437 KB window based application designed for concealing secret information in JPEG images. Its installation is easy can only be done by taking a few steps. It provides the functionality to conceal secret information into the working environment using the built-in browse function. SteganPEG also provides the feature of encryption by setting up passwords. This application is also able to calculate the size of concealed information to check

whether more information can be concealed or not. It is a lightweight application so would not burden computer performance [36].

rSteg 0404 This tool having a small size of 84 KB conceals text inside images, saves stego image to PNG format; decrypts the secret information by inserting a password. It provides the option to manually specify a convention text message that is to be encrypted and conceal it inside the image. It can be installed on all Windows versions having a Java working environment [37].

SSuite Picsel It is a free portable window-based 4.7 MB application to conceal text in an image. In this application text is concealed in an image and another image is used as a key to protecting concealed secret text. This means this tool provides a double level of security by applying for password protection first and then steganography. During extraction of secret information again required to enter key image [38].

Snow It is a free software program that conceals secret text information into tabs and space at the end of the lines. It is a command-line interface tool available for Unix/Linux platforms and its java version is also available. The manual page of the snow tool shows, “secret information is concealed into a text file by adding sequences of up to ‘7’ spaces, combined with tabs”, maximum of ‘3’ bits of secret information can be concealed for every ‘8’ columns in carrier file [39].

OurSecret 2.5 It is another shareware window-based 1.71 MB tool that is used to conceal secret information in a carrier file. Its interface is divided into two parts; the first interface conceal secret information in a carrier file and then another interface is used to reveal the secret information. Select carrier file to conceal secret information and type of secret information also. Enter the password for encryption of selected secret information and then conceal information. Use the same tool again to extract secret information [40].

QuickStego 1.2.0.1 It is an encryption tool constructed specially for concealing secret information into images to protect from unauthorized users. This tool permits users for uploading an image into the working environment using an integrated browser function. It conceals text into BMP, JPG, and GIF images. The final stego file will be in the BMP file format. During the testing phase of the application, it is found that the encryption procedure was very fast, and without any error [41].

HexaStego BMP 1.0 This tool conceals secret information into BMP images such that nobody doubts. It is a lightweight platform, which does not require installation to function. It can be stored in a USB (Universal Serial Bus) whenever need to practice it on other processors. For the working of this tool.net framework is required. After concealing secret information carrier file size increases

still, no visual changes occur. However, the main drawback of this tool is that only 50 KB of information can be concealed [42].

DeepSound 2.0 It is a steganography tool that can conceal text, image, and audio into Wav or FLAC (Free Lossless Audio Codec) audio files. This tool also provides password protection. Its installation is easy and adopts a window with an Explorer-based folder structure for navigating directories to trace audio files. It also offers password protection and good audio quality. This tool takes a short time to explore carrier files and then quickly conceal a secret file. When the cover audio file is converted to another format then many times this tool fails [43].

StegoStick It allows concealing of any type of secret information into any carrier file or any other file format such as PDF, EXE, CHM. This tool is created for text, audio, video, image steganography. It also provides encryption using these 4 types of encryption methods – RSA, DES, Triple DES, and AES [44].

DeEgger Embedder It is a user friendly and small software tool for concealing secret information into any digital media like AVI (Audio Video Interleave), JPG, PNG, MP3, Video, or other file formats. This tool allows choosing a host file where concealing secret information can be done. It is just like a game for a child, basically press on the “Combine” button and select the saving directory and filename to finish the task. For extracting concealed information, click on the feature included in the app. This tool does not provide features of compression, so because of the large size of the stego file suspension may arise to unauthorized users. Password protection is also not there in this tool so other users can simply access and change configuration settings [45].

JPHIDE and JPSEEK It is an image steganography tool used for concealing secret information inside a JPEG image. This tool can conceal secret information for up to ‘10%’. If large than ‘10%’ information is concealed then it can be detected by an unauthorized person. This tool is available freely for windows and DOS (Disk operating system) version. It is written in Visual C++ [46].

WbStego4 It is 32 bit open-source software for Windows. Any type of secret information can be concealed into carrier files such as Windows bitmaps up to 256 colors, ASCII or ANSI text files, HTML (Hypertext Markup Language) files, and Adobe PDF files. The information is concealed inside carrier files without modifying for unauthorized users. wbStego4 delivers two types of user interfaces. One is the wizard mode, which directs user step-by-step through encoding and decoding phases and the second is the flowchart mode for innovative users, to provide all settings in one time-saving overview flowchart. This tool conceals secret information in carrier files using the

following encryption algorithm: Blowfish, Twofish, CAST, and AES [47].

A summary of a few generally used, modern, open-source, freeware, and commercial steganography tools is presented in Table 1. For summarizing these tools details have been incorporated about types of secret information being concealed, techniques used to conceal secret information, the domain in which secret information is concealed, carrier digital media, stego media, encryption support, password protection, data compression, etc. Many of these tools stated can conceal secret information in a variety of digital media and different formats of the same media. Most of these tools provide compression, encryption, and password protection. These features make the

steganography tool more robust against various image processing attacks.

Document and text steganography techniques agree for variations in document structure and font layout to conceal secret information. For example, for HTML files, in terms of textual structure, modifying the source HTML is a form of text steganography. Moreover, the usage of both structural and semantic convertors delivers innovative opportunities.

Table 1 Comparison matrix of various steganography tools

Tool name	Concealed secret information	Carrier file type				Technique for concealing secret information	Additional features
		Image file	Audio file	Video file	Text file		
Steghide	Plain text	JPEG,BMP	Wav, AU	–	–	Graphic theoretical technique	Encryption, Hashing
SteganPEG	Any file type	JPEG	–	–	–	Partial decoding technique	Encryption, Password protection
OurSecret 2.5	Any file type	BMP, JPG, PNG	Wav	Mp4	–	–	Encryption
Snow	Plain text	–	–	–	Text	Whitespace	–
S-tool	Plain text	GIF, BMP	Wav			LSB, Large capacity of concealing	Encryption, Hashing, Compression
OpenStego	Plain text	JPEG, JPG, BMP, GIF, PNG	–	–	–	LSB Watermarking	Encryption
DeepSound 2.0	Plain text	JPEG	Wav, Flac	–	–	–	Encryption, Password protection
StegoStick	Plain Text	BMP	Wav	Mp4	PDF, EXE, CHM	–	Encryption
DeEgger Embedder	Any file type	AVI, JPG, PNG	Mp3	Mp4	–	–	Password protection
OpenPuff	Any file type	BMP, JPG, PCX, PNG, TGA	AIFF, MP3, WAV	3GP, MP4, MPG, VOB	PDF, SWF, FLV	DCT	Encryption, Scrambling, Whitening, Encoding
wbStego4	Any file type	JPEG, BMP	Wav	Mp4	ASCII, HTML, PDF	–	Encryption
and Seek	Any file type	BMP,GIF	Mp3	–	–	LSB	Cryptography, Password protection
VSL 1.1	Plain text	GIF	–	–	–	LSB, DCT	Classifiers SVM and BSM
Outguess-Rebirth	Plain text	JPEG	–	–	–	DCT	Cryptography
JSteg	Plain text	JPEG				LSB, DCT	Encryption, Compression
F5 Steganography	Plain text	JPEG	–	–	–	F5 DCT	Password protection

3 OpenPuff steganography tool

The comparison matrix in Table 1 suggests that the OpenPuff tool outperforms other tools in all the categories. Considering its efficiency, performance, flexibility, and reliability it is one of the most downloaded tools as observed in Fig. 2.

Referring to its popularity, a deep insight is done for the OpenPuff tool. An analysis of its performance, robustness, and efficiency has been done in the below sections on the parameters like concealing time, retrieval time, concealing capacity, histogram, visual inspection, etc. The salient features of OpenPuff tool that make it a preferred choice are, (1) it provides '4' layers of security which are encryption, scrambling, whitening, and encoding of concealed secret information [12], (2) this tool supports all digital media files in various formats and does not modify carrier file format upon concealing secret information [13]. Usually, steganography tools can conceal a small amount of secret information in a single carrier file. OpenPuff has the capability of concealing a large amount of secret information into multiple carrier files. Concealing secret information into multiple carrier files introduces randomness and improves the security of the tool [20]. Moreover, installation is not required to work with this tool so data like init and registry keys are not stored in a system that can provide trace information to hackers. Additionally, this tool is resistant against statistical and visual steganalysis which guarantees an advanced level of security while concealing secret information [12].

Many users recommended it as the best security tool. In Fig. 2 total downloads for OpenPuff demonstrate the popularity of the tool among the researchers and application users. The working of the OpenPuff steganography tool is shown in Fig. 3.

OpenPuff tool has a complex design but despite that enjoy a very good reputation with positive feedback from users. The user seems appreciating its ease of usability,

various features, and capability to handle multiple digital carriers [9].

3.1 Performance analysis of OpenPuff tool

Performance of OpenPuff tool has been analyzed based on features like concealing time, retrieval time, the security of concealed information, concealing capacity, histogram analysis, visual inspection analysis, etc. as explained below:

3.1.1 Concealing time

It is the time required to conceal the secret image in the carrier video frame. We used the Mp4 video as a carrier file and concealed the BMP image inside the carrier video frame. Information concealing time denotes the time taken to conceal secret information into the video frame 'f' in time interval 't' as shown in Eq. 1[4].

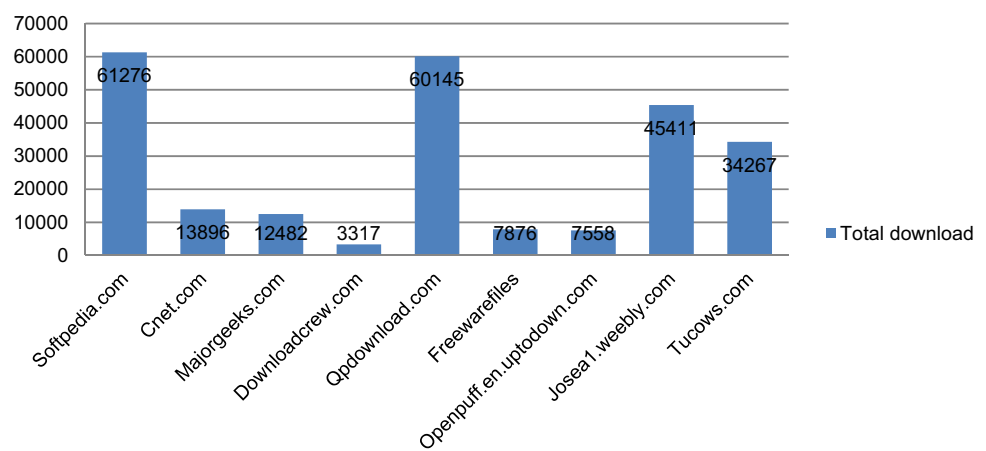
$$\text{Concealing time} = \frac{\text{Secret information} \times f}{t} \quad (1)$$

Concealing time varies with the size of secret information concealed and the format of carrier video taken. In Mp4 format, the average concealing time is 0.856 s, in VOB is 3.526 s whereas in MPG it extends to 19.321 s. In VOB and MPG format, concealing time is large because of the lossy compression of video format whereas in Mp4 it is very less due to lossless compression.

3.1.2 Retrieval time

It is the time taken to retrieve concealed secret information from the stego video at the time interval 't' and calculated as shown in Eq. 2 [20]. The time of information retrieval depends on the number of frames in the stego video. 'tot' denotes time taken to retrieve secret information.

Fig. 2 Total downloads for OpenPuff steganography tool as on 28-07-2020



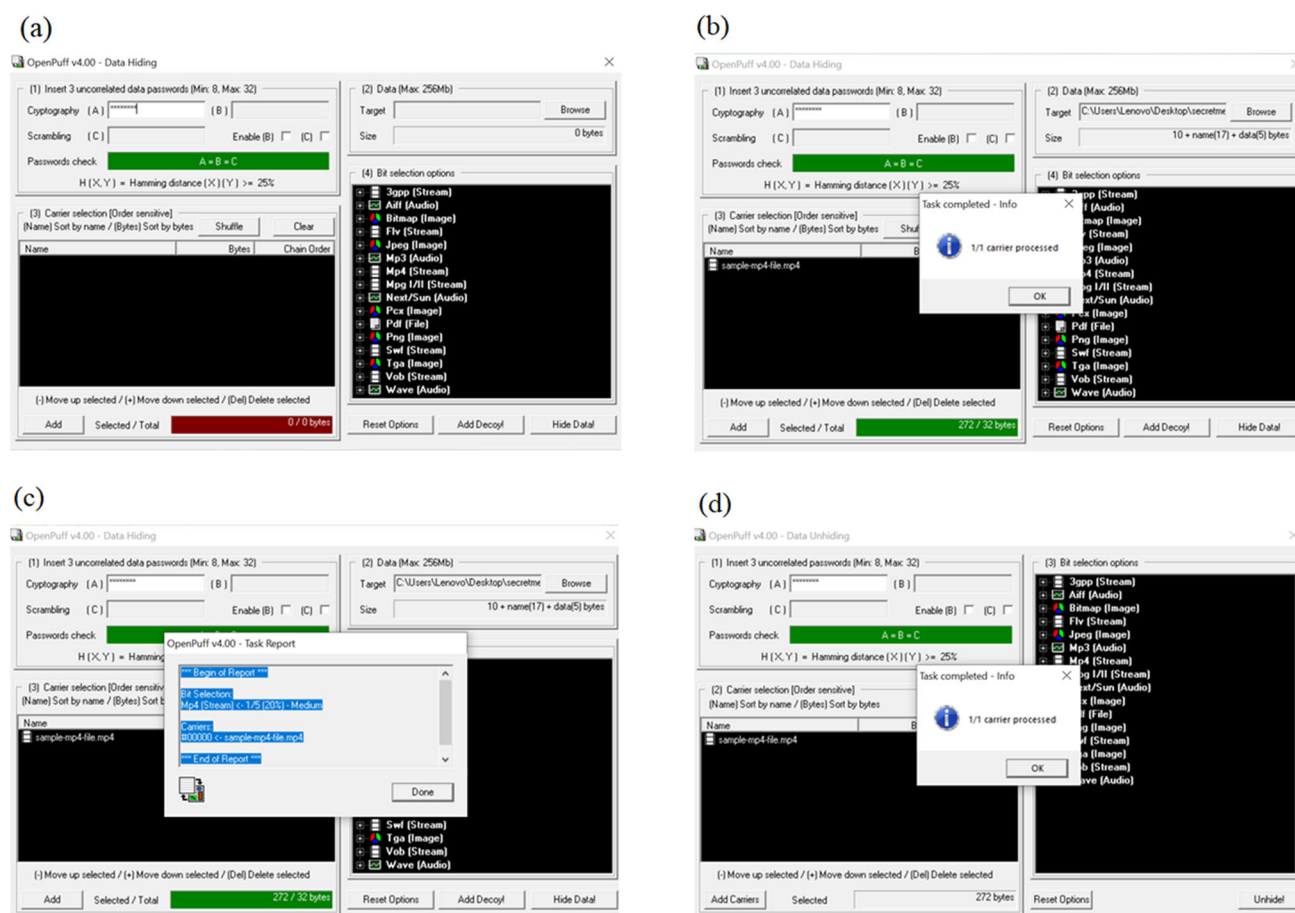


Fig. 3 Working of OpenPuff Tool a: password protection b: Concealing of the secret image in Mp4 video frame format c: Successfully concealed secret BMP image d: Extraction of the secret image

$$\text{Retrieval time} = \frac{\text{tot}(\text{secret information}) \times 100}{t} \quad (2)$$

In Mp4 video format, the retrieval time is smallest due to the least compressed format and largest in MPG. The average retrieval time determined is 0.552 s for Mp4, 1.778 s for VOB, and 6.34 s for MPG video format.

3.1.3 Security

Security can be measured based on the ratio of the difference between Concealed Information Sent (CIS) and Concealed Information Received (CIR) to concealed information received at the other side as shown in Eq. 3. A high difference between CIS and CIR means more security provided by the tool.

$$\text{Security} = \frac{(CIS - CIR)}{CIR} \quad (3)$$

The security of this tool is found to be 99.9% after calculation by using Eq. 3.

3.1.4 Concealing capacity

Concealing capacity means how much secret information can be concealed in frames of the Mp4 carrier video file and calculated using the formula given in Eq. 4 [5]. Mp4 video has the highest concealing capacity as compared to all other video formats like VOB, MPG, and 3GP, etc. Mp4 does not show any visual distortion when concealing the same amount of secret information whereas other formats like VOB, MPG, and 3GP, etc. can show some visual distortion.

$$\text{Concealing capacity} = \left(\frac{\text{Secret file size}}{\text{Carrier file size}} \times 100 \right) \quad (4)$$

Concealing capacity is calculated with the help of Eq. 4 and is found to be 57% of the carrier file.

3.1.5 Quality analysis for imperceptibility and robustness

Mainly there are two approaches through which image quality can be evaluated: The objective method and the Subjective method. The objective method is done through

mathematical comparisons among original and stego images whereas the Subjective method is the human review method that is not based on mathematical calculation. Histogram Analysis has been used as an objective method and Visual Human Review as a subjective method to review the imperceptibility and robustness respectively of the OpenPuff tool.

3.1.6 Histogram Analysis

For finding HA (Histogram Analysis), the experiment is done on Mp4 carrier video frames and after concealing secret BMP images inside the carrier frame. Histogram analysis can be done with the help of formula given in Eq. 5. It shows HA value between the histogram of the original Mp4 carrier video before and after concealing secret BMP image using concealing capacity '100%' of carrier video [3].

$$HA = \frac{\sum_{i=1}^n (His(c) - His(s))^2}{\sum_{i=1}^n His(c)^2} \quad (5)$$

where $His(c)$ and $His(s)$ are histograms of the carrier video frame and corresponding stego video frame respectively.

Histogram analysis is used to view the distribution of information in a file for checking whether information distribution change occurred from one time period to another. If there is slight or no change in information distribution of carrier video frame and stego video frame then it can be concluded that our steganography tool is robust. Carrier and stego video frames are shown in Fig. 4. By comparing the histogram of carrier and stego video frame in Fig. 5 it can be concluded that there is a negligible change that can be overlooked by naked human eyes.

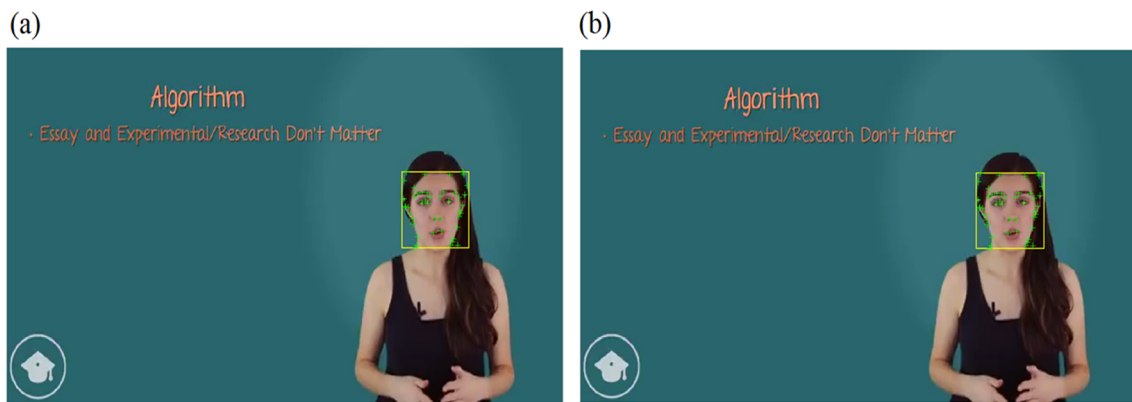


Fig. 4 Carrier v/s Stego Video Frame a: Carrier video frame b: Stego video frame

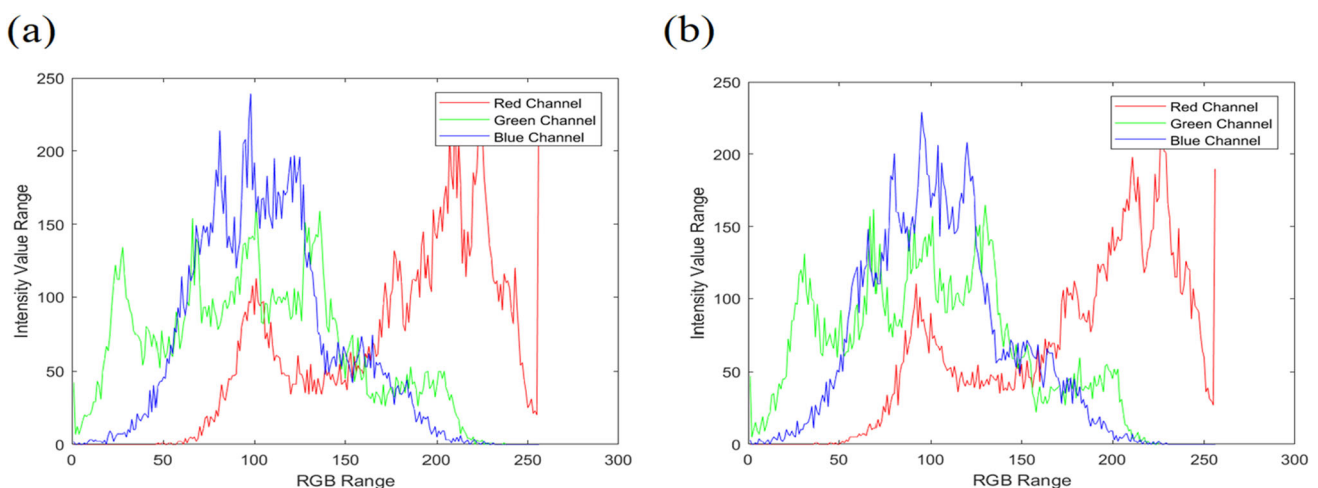
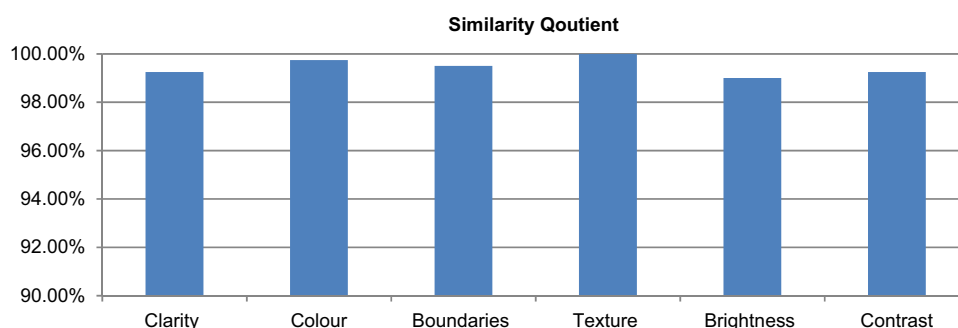


Fig. 5 Histogram comparison a: histogram of carrier video frame b: histogram of the stego video frame

Table 2 Reviews for carrier and stego video frame

Person	Age group	Clarity (%)	Colour (%)	Boundaries (%)	Texture (%)	Brightness (%)	Contrast (%)	Visual inspection review comment
Person1	15–25	100	100	100	100	100	100	Both are same images seeing the colour, brightness and contrast
Person2	15–25	100	100	100	100	100	100	No difference in colour, brightness. Everything is same
Person3	15–25	95	100	100	100	100	100	Clarity, contrast and colour are same, both the images are identical
Person4	15–25	100	95	100	100	100	95	Identical images, no differences in colour and texture
Person5	25–35	100	100	100	100	100	100	Not able to find any difference, both have same clarity
Person6	25–35	100	100	100	100	90	100	Same images in terms of colour, clarity and brightness
Person7	25–35	90	100	100	100	100	100	Brightness, colour, texture everything is same, seems to be same images
Person8	25–35	100	100	100	100	100	100	Images are identical in terms of brightness, color, and contract. No change in boundaries as well
Person9	35–45	100	100	100	100	100	90	Same colour, contract and clarity. No difference found
Person10	35–45	100	100	100	100	100	100	No difference in colour tone, brightness, contrast etc. Everything seems to be same
Average results	99.25%	99.75	99.50	100.00	99.00	99.25		

Fig. 6 Imperceptibility of Visual inspection

3.1.7 Visual Inspection Review

For testing the imperceptibility of carrier and stego video frame, both the video frames were reviewed with '10' persons of different age groups to check if they can figure out any differences with naked eyes in terms of clarity, color, boundaries, texture, brightness, contrast, etc. Carrier and stego video frames are shown in Fig. 4. The data gathered for visual inspection on the persons of different age groups are shown in Table 2. Figure 6 summarizes the similarity quotient of the cover file and stego file evaluated on some visual parameters like clarity, colour, boundaries, texture, etc.

Based on the survey results from '10' persons it has been concluded that both frames are found to be similar in all aspects including clarity, contrast, brightness, texture, color, and boundaries, etc. Visual inspection review proves the imperceptibility of the OpenPuff tool. After this survey, it has been proved that this tool maintains the visual quality of the stego frame which cannot be perceived through subjective (Visual Inspection Review) or objective methods (Histogram Analysis).

4 Conclusion

In this paper, a systematic study of the steganography tools developed in the last three decades has been done. The comparative analysis based on various features that include types of secret information being concealed, techniques used to conceal secret information, carrier digital media, stego media, encryption support, password protection, data compression, etc. gives an in-depth inspection of these tools. The efficiency and reliability of this tool are validated by analysis of the features concealing time, retrieval time, security of concealed information, concealing capacity, histogram analysis, and visual inspection analysis. The study has revealed that the tool supports all digital media files in various formats and doesn't modify carrier file format on concealing secret information. OpenPuff also has the capability of concealing a large amount of secret information into multiple carrier files. Concealing secret information into multiple carrier files introduces randomness and improves the security of the tool. This concludes why the OpenPuff steganography tool spawns a huge acceptance by academics and professionals.

Compliance with ethical standard

Conflicts of interest The authors have no competing interests in any form.

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